

Profile

Embedded systems engineer with experience spanning Embedded C/C++, FreeRTOS, ARM64 assembly, FPGA logic, serial interfaces, robotics, instrumentation, and field validation. Background includes low-level sensing/logging firmware, board bring-up, and C++/CUDA/OpenGL systems tools.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed Embedded C and FreeRTOS functionality for geological well tools using Lattice FPGA, PIC32, dsPIC, ESP32, STM32, and C8051-class hardware platforms.
- Implemented sensor-logging and control-side logic and firmware for field-deployed systems operating in instrumentation-heavy environments.
- Performed oscilloscope-based debugging, bench validation, and field testing to verify hardware-software integration and system behavior.
- Supported selective PCB updates in Altium and related board-level engineering tasks as system requirements evolved.
- Built C++/CUDA host-side applications for waveform processing, visualization, and diagnostics around the embedded stack; created C++ GUI applications with OpenGL custom plotting controls for large sensor datasets.
- Implemented a supporting C#/ASP.NET web server and Microsoft SQL Server services for remote tool calibration and data collection.

Yale University

Embedded Systems Teaching Assistant / PhD Researcher

2021–present

- Supported embedded systems laboratories and student projects involving firmware, hardware interfaces, debugging practices, and instrumentation workflows.
- Continued broader engineering work in scientific computing and simulation while maintaining strong embedded-systems fluency.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026 on excited-state materials physics.

Selected Projects

EEG-Driven Robotic Hand

- Built a PIC32-based robotic-hand controller driven by EEG sensor input, integrating biosignal acquisition, embedded control logic, and servo actuation.

PIC32 / Quartus Maze-Mapping Robot

- Built a PIC32 robot that mapped a maze and transmitted the map for remote visualization; used an FPGA programmed in Quartus to support mapping and display workflows.

Single-Cycle MIPS Processor

- Implemented a small Verilog CPU and self-checking test bench to demonstrate digital design, datapath organization, and reproducible verification.

Matrix-Multiply Accelerator

- Built a compact Verilog accelerator block with clear start/done control behavior and a simulation-backed test flow.

Host-Side Diagnostic and Visualization Tooling

- Developed CUDA-based utilities for analyzing and visualizing waveform data associated with embedded sensing and logging systems.

Education

Yale University

PhD, Electrical Engineering

2021–present

Ongoing doctoral work; also served as an embedded systems teaching assistant.

Yale University

MPhil, Electrical Engineering

2023

Yale University*MS, Electrical Engineering*

2023

Cornell University*MEng, Electrical and Computer Engineering*

2017–2018

Included advanced engineering coursework and embedded-adjacent systems work.

Cornell University*BS, Electrical and Computer Engineering*

2013–2017

Included embedded systems, robotics, and hardware-software integration foundations.

Selected Coursework

Embedded: Advanced Embedded Systems; laboratory-oriented firmware and systems work**Signals and Systems:** Mathematics of Signals and Systems**Hardware:** VLSI Design; Analog VLSI Design; digital and mixed engineering foundations**Supporting Theory:** Semiconductor Physics; controls/robotics-adjacent systems exposure in ECE training

Technical Skills

Languages: Embedded C/C++, ARM64 assembly, Python, CUDA, Bash**Firmware / Hardware:** FreeRTOS, Verilog, Lattice FPGA, Quartus, PIC32, dsPIC, ESP32, STM32, C8051, sensor logging, embedded control**Serial Interfaces:** SPI, I2C, UART/USART, serial device control, hardware-software integration workflows**Validation:** Oscilloscopes, bench instrumentation, field testing, board bring-up, Altium update support**Systems Software:** Linux, C++ GUI, CUDA, OpenGL custom plotting controls, C#; C#/ASP.NET and Microsoft SQL Server for internal calibration services